Introduction to Object Oriented Programming :→

Real World Entity : Student / Teacher / Pen / Pencil

Student → Characteristics :
- name
- age
- USN
- email
- department

} attributes / properties / characteristics / instance variables

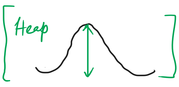
```
Class Student {
    String name;
    int age;
    int USN;
    String email;
    String department;
}
```

Student s1 = new Student();
s1.→ (all data access)

Class : A class is a blueprint / template / prototype to represent real world entities which are called (objects / references / instances) in programming.
A class has no memory of its own. It belongs to the static memory.

Object : An object is a real world entity or an instance of a class. It is also a key or reference to access all the data inside the class.

* An object / instance / reference has memory of its own. All objects belong to the Heap Memory of your OS.
* More objects ==== Increasing Heap



* It is highly recommended not to create unnecessary objects.
* Unnecessary objects will break / reduce the efficiency of your application.

Methods : Speak( ); write( ); read( );

In case of Java, functions are called methods because they describe the behaviour of an object.

(DRY: Don't Repeat Yourself) (Avoid Redundancy)

Constructor : It is a special type of method used to instantiate / initialize / create objects of a class.

Points to remember :

(i) A constructor is the same name as the class followed by a parantheses → ClassName();
(ii) A constructor has no return type.
(iii) A constructor should be either public or default → otherwise objects cannot be created.
(iv) There can be multiple constructors inside a class → Constructor Overloading (Polymorphism)
(v) There are two types of constructors:
  (a) Default Constructor / No-argument constructor
  (b) Parameterised Constructor
(vi) If the user doesn't create the constructor, the compiler (JVM) generates the default constructor to initialize objects.
(vii) When we create our own parameterised constructor, the default constructor gets destroyed or deleted.

* this keyword is used for accessing current class data members (variables, methods, etc)
* this keyword is used to remove ambiguity or confusion while there exists duplicate names.
* Static is used for common parameters at class level.
* In C++ objects need to be destroyed by using destructor. ~ClassName();
* In Java, garbage Collection & Clutter Deletion is handled & managed by the JVM.

* How do we copy an object into another ?
→ { Copy Constructor } ⇒ C++ { Not available in Java }

* In Java we use custom copy methods which are static, to copy one object into another.
* We should not do this: → Class obj2 = obj1;
(It can lead to redundancy & errors)
⇓
Unknown Exceptions